

When using doxorubicin as part of chemotherapy regimens which combine drugs of similar pharmacological effects (i.e. cytotoxicity) additive toxicity is likely to occur. Such additive toxicity has to be taken into consideration especially with regard to bone marrow function. Doxorubicin is a powerful but temporary immunosuppressant agent. Appropriate measures should be taken to prevent secondary infection. Clinical consequences of severe myelosuppression include fever, infection, sepsis/septicaemia, septic shock, haemorrhage, tissue hypoxia or death.

Secondary Leukaemia

Secondary leukaemia, with or without a pre-leukaemic phase, has been reported in patients treated with anthracyclines including doxorubicin. Secondary leukaemia is more common when such drugs are given in combination with DNA-damaging antineoplastic agents, in combination with radiotherapy, when patients have been heavily pre-treated with cytotoxic drugs, or when doses of the anthracyclines have been escalated. This has been noted in the adjuvant and neoadjuvant setting. These leukaemias can have a 1- to 3-year latency period.

Gastrointestinal

Doxorubicin is emetogenic. Mucositis is a frequent and painful complication of doxorubicin treatment but is less common than myelosuppression. Mucositis/stomatitis generally appears early after drug administration, most commonly developing 5 to 10 days after treatment. It typically begins as a burning sensation in the mouth and pharynx. The mucositis may involve the vagina, rectum and oesophagus, and, if severe, may progress over a few days to mucosal ulcerations with risk of secondary infection. Most patients recover from this adverse event by the third week of therapy. Retrospective comparison of the incidence of mucositis suggests that it is less frequent as the intervals between doses increase. Mucositis may be severe in patients who have had previous irradiation to the mucosae.

Obesity

The systemic clearance of doxorubicin has been found to be reduced in obese patients; such patients have to be carefully monitored if undergoing treatment with the maximum recommended doses of the drug.

Effects at Site of Injection

Phlebosclerosis may result from an injection into a small vessel or from repeated injections into the same vein. Following the recommended administration procedures may minimise the risk of phlebitis/thrombophlebitis at the injection site.

Extravasation

Extravasation of doxorubicin during intravenous injection may produce local pain (a burning or stinging sensation), severe tissue lesions (vesication, severe cellulitis) and necrosis. Should signs or symptoms of extravasation occur during intravenous administration of doxorubicin, the drug infusion should be immediately stopped.

To minimise perivenous infiltration.

Tumour-Lysis Syndrome

Like other cytotoxic drugs, doxorubicin may induce hyperuricaemia as a consequence of the extensive purine catabolism that accompanies rapid drug-induced lysis of neoplastic cells (tumour-lysis syndrome). The clinician should monitor the patient's blood uric acid level, potassium, calcium phosphate and creatinine. Supportive and pharmacologic measures should be used to control this problem. Hydration, urine alkalisation and prophylaxis with allopurinol to prevent hyperuricaemia may minimise potential complications of tumour-lysis syndrome.

Immunosuppressed Effects/Increased Susceptibility to Infections

Administration of live or live-attenuated vaccines in patients immunocompromised by chemotherapeutic agents, including doxorubicin, may result in serious or fatal infections. Vaccination with a live vaccine should be avoided in patients receiving doxorubicin. Killed or inactivated vaccines may be administered; however, the response to such vaccines may be diminished.

Other

Doxorubicin may potentiate the toxicity of other anticancer therapies. Exacerbation of cyclophosphamide induced haemorrhagic cystitis, mucositis induced by radiotherapy, and enhanced hepatotoxicity of 6-mercaptopurine have been reported. Radiation-induced toxicities (myocardium, mucosae, skin and liver) have also been reported.

As with other cytotoxic agents, thrombophlebitis and thromboembolic phenomena, including pulmonary embolism (in some cases fatal), have been coincidentally reported with the use of doxorubicin. Doxorubicin imparts a red colouration to the urine for 1-2 days after administration and patients should be advised to expect this during active therapy.

Intravesical Route

Administration of doxorubicin by the intravesical route may produce symptoms of chemical cystitis (such as dysuria, polyuria, nocturia, stranguria, haematuria, bladder discomfort, necrosis of the bladder wall) and bladder constriction. Special attention is required for catheterisation problems (e.g., urethral obstruction due to massive intravesical tumours). Urine cytologies and blood counts should be monitored monthly and cytoscopic examinations should be performed at regular intervals.

Instructions to be Given to Patients

1. Patients should inform their physicians immediately if pain develops at the infusion site.

2. Nausea and vomiting may be expected 3-6 hours after drug treatment, and may last for several hours.

3. Patients should be advised to expect a red colouration to the urine (not indicative of haematuria) for 1 to 2 days after each administration of Doxorubicin Concentrate for solution for infusion.

4. Alopecia (hair loss) should be expected 1 to 2 weeks after the initiation of Doxorubicin Concentrate for solution for infusion treatment. Hair loss may be complete but hair always returns after termination of treatment.

NB. Scalp tourniquets inflated to above systolic blood pressure and left in situ for 30 minutes over the time of Doxorubicin Concentrate for solution for infusion treatment reduces the probability of alopecia.

5. Anorexia may be expected for 24 hours following each treatment and occasionally may persist for several days.

6. Hyperpigmentation, usually in the hands, nails and buccal mucosa may develop in patients receiving Doxorubicin Concentrate for solution for infusion. Patients should be advised that this condition does not usually improve after termination of treatment.

7. Infertility in both sexes is usual in patients receiving Doxorubicin Concentrate for solution for infusion. Amenorrhoea is frequent and in pre-menopausal women, regular menstruation usually returns a few months after termination of Doxorubicin Concentrate for solution for infusion therapy. This is often accompanied by normal fertility.

8. Male patients should be advised that oligospermia or azoospermia may be permanent. There is a possibility that fertility may return several years after ceasing therapy. Men undergoing Doxorubicin Concentrate for solution for infusion therapy should be advised to use effective contraceptive measures.

9. Patients should be instructed to inform their physicians of any prior abnormal heart or liver conditions, as this information is vital to the formulation of appropriate dosage regimes.

Use in Hepatic Impairment

The major route of elimination of doxorubicin is the hepatobiliary system. Toxicity to recommended doses of doxorubicin is enhanced by hepatic impairment. Therefore, prior to the individual dosing, evaluation of hepatic function is recommended using conventional clinical laboratory tests such as AST (SGOT), ALT (SGPT), alkaline phosphatase, bilirubin and BSP. Serum total bilirubin levels should also be evaluated during treatment with doxorubicin. Patients with elevated bilirubin may experience slower clearance of drug with an increase in overall toxicity. Lower doses are recommended in these patients. Patients with severe hepatic impairment should not receive doxorubicin.

Changes in hepatic function induced by concomitant therapies, either given to achieve optimal antitumour efficacy or given for the pharmacological management of concomitant diseases, may affect doxorubicin metabolism, pharmacokinetics, therapeutic efficacy or toxicity

Use in the Elderly

No data available.

Paediatric Use

No data available.

Effects on Laboratory Tests

No data available

'IAUHKAN UBAT DARIPADA CAPAIAN KANAK-KANAK'

KEEPOUT OF REACH OF CHILDREN

Please consult your pharmacist / doctor before taking this product.

m) Interaction with other medicaments

Doxorubicin is a major substrate of cytochrome P450 CYP3A4 and CYP2D6, and P-glycoprotein (P-gp). Clinically significant interactions have been reported with inhibitors of CYP3A4, CYP2D6, and/or P-gp (e.g. verapamil), resulting in increased concentration and clinical effect of doxorubicin. Inducers of CYP3A4 (e.g. phenobarbital, phenytoin, St. John's Wort) and P-gp inducers may decrease the concentration of doxorubicin.

Doxorubicin is mainly used in combination with other cytotoxic agents. Additive toxicity may occur, especially with regard to bone marrow/haematologic and gastrointestinal effects.

Adjuvant Chemotherapy Involving Doxorubicin

It is not recommended that doxorubicin be used routinely as adjuvant chemotherapy in any tumour category. The activity of doxorubicin in combination with other drugs is affected not only by the nature of the drug itself, but also by the schedule of administration. It is strongly recommended that in situations where doxorubicin is intended for use as adjuvant chemotherapy, higher authorities as well as the Hospital Ethical Committee be consulted.

Cyclophosphamide

Concurrent cyclophosphamide treatment sensitises the heart to the cardiotoxic effects of doxorubicin. Doxorubicin may exacerbate cyclophosphamide cystitis.

Cyclosporin

The addition of cyclosporin to doxorubicin may result in increases in area under the concentration-time curve (AUC) for both doxorubicin and doxorubicinol, possibly due to a decrease in clearance of the parent drug and a decrease in metabolism of doxorubicinol. Literature reports suggest that adding cyclosporin to doxorubicin results in more profound and prolonged hematologic toxicity than that observed with doxorubicin alone. Coma and seizures have also been described with concomitant administration of cyclosporin and doxorubicin.

Heparin

Doxorubicin should not be mixed with heparin since it has been reported that these drugs are incompatible to the extent that a precipitate may form.

Mediastinal Radiotherapy

Concurrent mediastinal radiotherapy and doxorubicin may be associated with enhanced myocardial toxicity of doxorubicin.

Paclitaxel:

Paclitaxel can cause increased plasma-concentration of doxorubicin and/or its metabolites when given prior to doxorubicin. Certain data indicate that this effect is minor when anthracycline is administered prior to paclitaxel.

Propranolol

In view of the finding that doxorubicin and propranolol have both been shown to inhibit cardiac mitochondrial CoQ10 enzymes it is possible that such a drug interaction may result in an additive cardiotoxic effect.

Radiotherapy

Concurrent radiotherapy and doxorubicin treatment may be associated with increased radiation toxicity, i.e., skin reactions and mucositis.

Sorafenib:

Both increases (21% - 47%) and no change in the AUC of doxorubicin were observed with concomitant treatment with sorafenib 400 mg twice daily. The clinical significance of these findings is unknown.

n) Fertility, Pregnancy and lactation

Effects on Fertility

Doxorubicin may cause infertility during the time of drug administration. In women, doxorubicin may cause amenorrhea. Although ovulation and menstruation appear to return after termination of therapy, premature menopause can occur.

Doxorubicin was toxic to male reproductive organs in animal studies, producing testicular atrophy, diffuse degeneration of the seminiferous tubules, and hypospermia.

Doxorubicin is mutagenic and can induce chromosomal damage in human spermatozoa. Oligospermia or azospermia may be permanent; however, sperm counts have been reported to return to normospermic levels in some instances. This may occur several years after the end of therapy.

Men undergoing treatment with doxorubicin should use effective contraceptive measures.

Use in Pregnancy - Category D

There is no information on the drug's use in pregnancy; therefore, the drug should not be used in pregnant women or those likely to become pregnant unless the expected benefit outweighs any potential risk. If a woman receives doxorubicin during pregnancy or becomes pregnant while taking the drug, she should be apprised of the potential hazard to the fetus.

Although animal studies have not demonstrated teratogenic activity due to doxorubicin treatment, an embryotoxic action is evident. Studies with rabbits and rats have revealed a decreased weight gain and a higher incidence of resorbed fetuses. No greater incidence of gross, visceral or skeletal malformations or of post-natal deaths has been observed.

Dose-related mutagenic effects of doxorubicin have been reported to produce severe chromosomal aberrations in in vitro studies. In view of this activity, the use of this drug in pregnant women is not recommended.

Use in Lactation

Doxorubicin is secreted in breast milk. Women should be instructed not to breast-feed while undergoing treatment with doxorubicin.

o) Side effects

Adverse reactions reported in association with doxorubicin therapy are listed below by MedDRA System Organ Class and by frequency. Frequencies are defined as: Very common (≥10%), Common (≥1%, <10%), Uncommon (≥0.1%, <1%), Rare (≥0.01%, <0.1%), Very rare (<0.01%), and Not known (cannot be estimated from available data).

Table 3: Adverse Reactions

Blood and Lymphatic System Disorders		Investigations	
Very common	Leucopenia, neutropenia, anaemia, thrombocytopenia	Very common	Decreased ejection fraction, abnormal electrocardiogram, abnormal transaminases, increased weightlb
Cardiac Disorders		Metabolism and Nutrition Disorders	
Common	Cardiomyopathy, congestive cardiac failure, sinus tachycardia	Very common	Decreased appetite
Not known	Atrioventricular block, Tachyarrhythmia, Bundle branch block, pericardial effusion.	Not known	Dehydration, Hyperuricaemia
Eye Disorders		Neoplasms Benign, Malignant and Unspecified (including cysts and polyps)	
Common	Conjunctivitis	Not known	Acute lymphocytic, leukaemia, acute myeloid leukaemia
Not known	Keratitis, increased Lacrimation	Nervous System Disorders	
Gastrointestinal Disorders		Not known	Somnolence
Very common	Mucosal inflammation/stomatitis, diarrhoea, vomiting, nausea	Renal and Urinary Disorders	
Common	Oesophagitis, Abdominal pain	Not known	Chromaturia, renal disorder
Not known	Gastrointestinal haemorrhage, erosive gastritis, colitis, mucosal discolouration, large intestinal haemorrhage, gastrointestinal necrosis, large intestinal ulcer	Reproductive System and Breast Disorders	
General Disorders and Administration Site Conditions		Not known	Amenorrhoea, Azoospermia, Oligospermia
Very common	Pyrexia, asthenia, chills	Skin and Subcutaneous Tissue Disorders	
Common	Infusion site reaction, extravasation	Very common	Palmar-plantar erythrodysaesthesia syndrome, Alopecia
Not known	Malaise, injection site erythema, death	Common	Urticaria, Rash, Skin hyperpigmentation, Nail hyperpigmentation
Immune System Disorders		Not known	Recall phenomenon Photosensitivity reaction, , Pruritus, Skin disorder, skin necrosis, skin wrinkling, blister
Not known	Anaphylactic reaction	Vascular Disorders	
Infections and infestations		Uncommon	Embolism
Very common	Infection	Not known	Shock, haemorrhage, Thrombophlebitis, Phlebitis, Phlebosclerosis, hot flush, flushing
Common	Sepsis, cellulitis		
a. Reported in patients with acute myelogenous leukaemia treated with combination doxorubicin and cytarabine			
b. Reported in patients with early breast cancer receiving doxorubicin-containing adjuvant therapy.			
c. For one to two days after administration.			

Intravesical use

Systemic toxicity is not a common problem, however, adverse reactions have been noted at doses exceeding that recommended.

Local reactions observed include chemical cystitis, contraction of the bladder, haematuria, painful micturition, frequency and urgency. These disturbances are transient.

p) Symptoms and treatment of over dose

Acute overdose with doxorubicin will result in acute cardiac alterations, severe myelosuppression (mainly leucopenia and thrombocytopenia), and gastrointestinal toxic effects (mainly mucositis).

Delayed cardiac failure may occur up to six months after the overdose. Patients should be observed carefully and should signs of cardiac failure arise, be treated along conventional lines.

Single doses of 250 mg and 500 mg of doxorubicin have proved fatal. Such doses may cause acute myocardial degeneration within 24 hours and severe myelosuppression, the effects of which are greatest between 10 and 15 days after administration.

Toxic blood levels have not been established. Doxorubicin is highly protein bound; however, if haemoperfusion is initiated within minutes of an overdose, a reduction in serum levels can be achieved. Haemodialysis is unlikely to be effective.

There is no specific antidote for doxorubicin. Symptomatic supportive measures should be instituted. Support respiratory and cardiac function. Cardiac monitoring is recommended. Particular attention should be given to prevention and treatment of possible severe haemorrhages or infections secondary to severe, persistent bone marrow depression.

q) Effects on Ability to drive and use Machine

The effects of this medicine on a person's ability to drive and use machines were not assessed as part of its registration.

r) Preclinical Safety Data

Not Applicable

s) Instructions for use.

Dosage:

Care in the administration of Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL will reduce the chance of perivenous infiltration. It may also decrease the chance of local reactions such as urticaria and erythematous streaking.

The recommended dosage schedule is 60-75 mg/m² as a single intravenous injection administered at 21-day intervals. The lower dose should be given to patients with inadequate marrow reserves due to old age, or prior therapy, or neoplastic marrow infiltration.

An alternative dose schedule is 30 mg/m² on each of three successive days repeated every 4 weeks. The adult dosage regimens may be suitable for paediatric cases.

The recommended lifetime cumulative dose limit is 550 mg doxorubicin/m² body surface area. Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL has been administered as an intra-arterial infusion for 1-3 days at doses of 45-100 mg/m². It is recommended that the total cumulative dose of doxorubicin for adults aged 70 or older be restricted to 450 mg/m² body surface area.

Method of Administration

Doxorubicin Concentrate for solution for infusion, USP 2mg/mL must be handled with care. If contact with the skin occurs, wash thoroughly with soap and water.

Protective measures

The following protective recommendations are given due to the toxic nature of this substance:

- Personnel should be trained in good technique for reconstitution and handling.
- Pregnant staff should be excluded from working with this drug.
- Personnel handling doxorubicin should wear protective clothing: goggles, gowns and disposable gloves and masks.
- A designated area should be defined for reconstitution (preferably under a laminar flow system). The work surface should be protected by disposable, plastic-backed, absorbent paper.
- All items used for reconstitution, administration or cleaning, including gloves, should be placed in high-risk waste-disposal bags for high-temperature incineration.
- Spillage or leakage should be treated with dilute sodium hypochlorite (1% available chlorine) solution, preferably by soaking, and then water.
- All cleaning materials should be disposed of as indicated previously.
- In case of skin contact thoroughly wash the affected area with soap and water or sodium bicarbonate solution. However, do not abrade the skin by using a scrub brush.
- In case of contact with the eye(s), hold back the eyelid(s) and flush the affected eye(s) with copious amounts of water for at least 15 minutes. Then seek medical evaluation by a physician.
- Always wash hands after removing gloves.

Intravenous Infusion

It is recommended that Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL be slowly administered into the tubing of a freely running intravenous infusion of Sodium Chloride Solution for Injection or 5% Glucose solution for Injection. The tubing should be attached to a butterfly needle inserted preferably into a large vein. The rate of administration is dependent on the size of the vein and the dosage. However, the dose should be administered in not less than 3-5 minutes. A direct push injection is not recommended due to the risk of extravasation, which may occur even in the presence of adequate blood return upon needle aspiration.

Local erythematous streaking along the vein as well as facial flushing may be indicative of too rapid administration. A burning or stinging sensation may be indicative of perivenous infiltration and the infusion should be immediately terminated and restarted in another vein.

Intravesical administration

The following procedure is recommended:

1. The bladder should be catheterised and emptied.
2. Dilute Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL to a final concentration of 80 mg in 100 mL of normal saline and instil via the catheter into the bladder.
3. The catheter should be removed and the patient instructed to be on one side. At 15-minute intervals, the patient should alternate to the opposite side over a 1-hour period.
4. The patient should be requested not to urinate for 1 hour, after which the bladder should be emptied of solution.
5. The procedure should be repeated at monthly intervals.

Until specific compatibility data are available, it is not recommended that Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL be mixed with other drugs. Contact with alkaline solutions should be avoided since this can lead to hydrolysis of doxorubicin. Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL should not be mixed with heparin due to chemical incompatibility that may lead to precipitation.

Doxorubicin should not be mixed with fluorouracil (e.g., in the same IV infusion bag or at the Y-site of an IV infusion line) since it has been reported that these drugs are incompatible to the extent that a precipitate might form. If concomitant therapy with doxorubicin and fluorouracil is required, it is recommended that the IV line be flushed between the administration of these drugs. Also see Section **Incompatibilities**.

Doxorubicin Concentrate for Solution for Infusion, USP 2mg/mL has been used in combination with other approved chemotherapeutic agents.

Though evidence is available that at least in some type of neoplastic disease combination chemotherapy is superior to single agents the benefits and risks of such therapy have not yet been fully elucidated.

Dosage adjustments

Use in hepatic impairment

Doxorubicin dosage must be reduced if hepatic function is impaired according to the following table:

Table 4: Recommended dose for patients with hepatic impairment

Serum Bilirubin Levels	BSP Retention	Recommended Dose
20 - 50 µmol/L	9 - 15%	1/2 normal dose
>50 µmol/L	>15%	1/4 normal dose

Doxorubicin should not be administered to patients with severe hepatic impairment.

Use in the elderly:

A lower dose may need to be considered in elderly patients with inadequate marrow reserves due to old age. It is recommended that the total cumulative dose of doxorubicin for adults aged 70 or older be restricted to 450 mg/m² body surface area.

Other special populations

A lower doses or longer intervals between cycles may need to be considered for pretreated patients, obese patients, or patients with neoplastic bone marrow infiltration.

t) Storage conditions

Store at 2°C to 8°C (refrigerate, do not freeze). Protect from light.

After dilution with 0.9% sodium chloride solution and 5% glucose solution, the diluted product to be infused within an hour at room temperature.

Nature and contents of container

10 mg/5mL: Clear, red color solution free from visible particles in 5 mL Tubular clear 20 mm collar flat bottom glass vial along with 20 mm grey fluorotec coated chlorobutyl rubber stopper and sealed with 20 mm Aluminium flip-off seal of lemon yellow color button packed along with package insert.

1 vial packed in a carton along with package insert.

50 mg/25 mL: Clear, red color solution free from visible particles in 30 mL Tubular clear 20 mm collar flat bottom glass vial along with 20 mm grey fluorotec coated chlorobutyl rubber stopper and sealed with 20 mm Aluminium flip-off seal of lemon yellow color button packed along with package insert.

1 vial packed in a carton along with package insert.

u) Shelf life

The shelf life of unopened vials is 24 months.

After dilution with 0.9% sodium chloride solution and 5% glucose solution, the diluted product to be infused within an hour at room temperature.

v) Therapeutic code: ATC Code: L01DB01

w) Name and address of manufacturer/ product registration holder

Manufactured by:

 **SPACURE labs Pvt.Ltd.**
Plot No.12, Biotech Park, Phase 2, Lalgadda (Malakpet) (V),
Shameerpet (M), Medchal (Dist), Telangana (State),
500101, India.
Product Registration Holder
Healol Pharmaceuticals Sdn.Bhd.,
74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju,
53300 Kuala Lumpur, Malaysia.

x) Date of Revision of PI:
08/01/2024